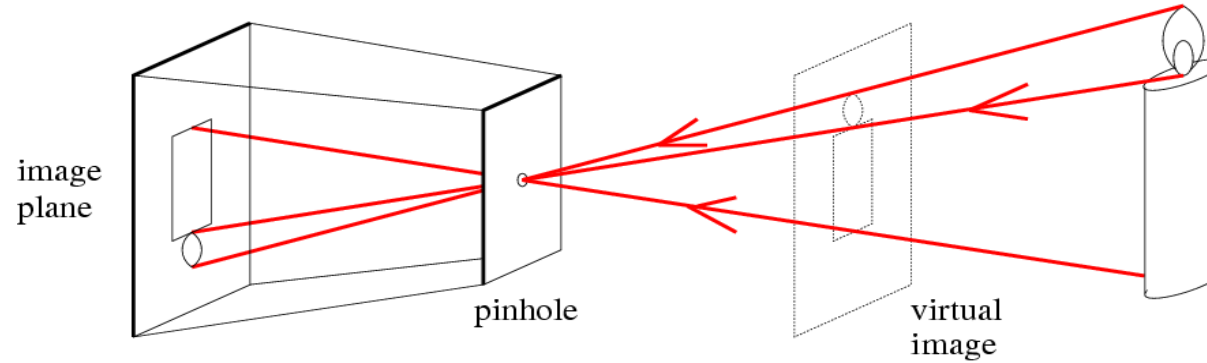


Lecture 1.3

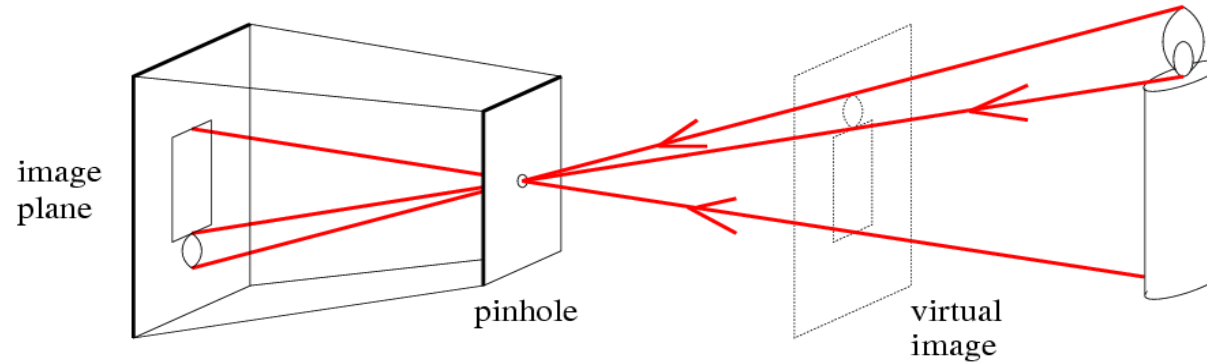
Basic projective geometry

Motivation



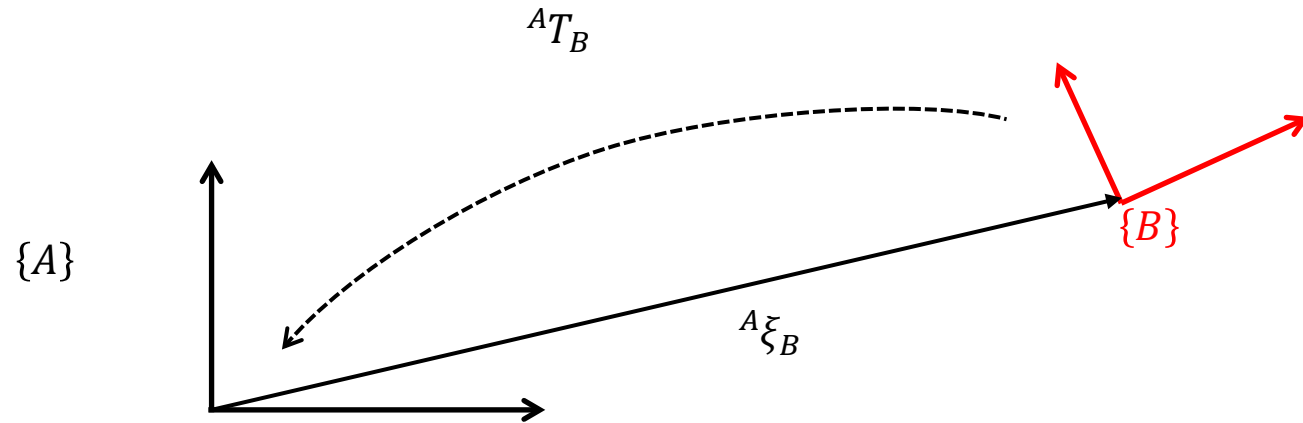
- For the pinhole camera, the correspondence between observed 3D points in the world and 2D points in the captured image is given by straight lines through a common point (pinhole)
- This correspondence can be described by a mathematical model known as “*the perspective camera model*” or “*the pinhole camera model*”
- This model can be used to describe the imaging geometry of many modern cameras, hence it plays a central part in computer vision

Motivation



- Before we can study the perspective camera model in detail, we need to expand our mathematical toolbox
- We need to be able to mathematically describe the position and orientation of the camera relative to the world coordinate frame
- Also we need to get familiar with some basic elements of projective geometry, since this will make it MUCH easier to describe and work with the perspective camera model

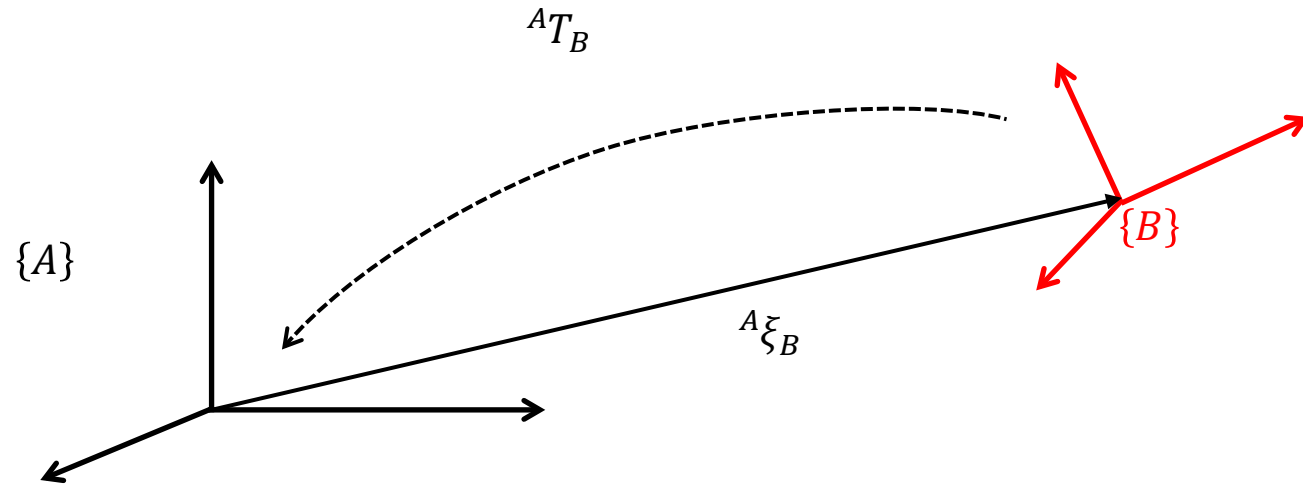
Introduction



- We have seen that the pose of a coordinate frame $\{B\}$ relative to a coordinate frame $\{A\}$, denoted ${}^A\xi_B$, can be represented as a homogeneous transformation AT_B in 2D

$${}^A\xi_B \mapsto {}^AT_B = \begin{bmatrix} {}^AR_B & {}^A\mathbf{t}_B \\ \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & {}^At_{Bx} \\ r_{21} & r_{22} & {}^At_{By} \\ 0 & 0 & 1 \end{bmatrix} \in SE(2)$$

Introduction



- We have seen that the pose of a coordinate frame $\{B\}$ relative to a coordinate frame $\{A\}$, denoted ${}^A\xi_B$, can be represented as a homogeneous transformation AT_B in 2D and 3D

$${}^A\xi_B \mapsto {}^AT_B = \begin{bmatrix} {}^AR_B & {}^A\mathbf{t}_B \\ \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & {}^At_{Bx} \\ r_{21} & r_{22} & r_{23} & {}^At_{By} \\ r_{31} & r_{32} & r_{33} & {}^At_{Bz} \\ 0 & 0 & 0 & 1 \end{bmatrix} \in SE(3)$$

Introduction

- And we have seen how they can transform points from one reference frame to another if we represent points in homogeneous coordinates

$$\mathbf{p} = \begin{bmatrix} x \\ y \end{bmatrix} \mapsto \tilde{\mathbf{p}} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \qquad \mathbf{p} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mapsto \tilde{\mathbf{p}} = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

- The main reason for representing pose as homogeneous transformations, was the nice algebraic properties that came with the representation

Introduction

- Euclidean geometry

- ${}^A\xi_B \mapsto ({}^A R_B, {}^A \mathbf{t}_B)$
- Complicated algebra

$$\begin{aligned} {}^A \mathbf{p} &= {}^A \xi_B \cdot {}^B \mathbf{p} &\mapsto & {}^A \mathbf{p} = {}^A R_B {}^B \mathbf{p} + {}^A \mathbf{t}_B \\ {}^A \xi_C &= {}^A \xi_B \oplus {}^B \xi_C &\mapsto & ({}^A R_C, {}^A \mathbf{t}_C) = ({}^A R_B {}^B R_C, {}^A R_B {}^B \mathbf{t}_C + {}^A \mathbf{t}_B) \\ \ominus {}^A \xi_B &&\mapsto & ({}^A R_C^T, -{}^A R_C^T {}^A \mathbf{t}_C) \end{aligned}$$

- Projective geometry

- ${}^A \xi_B \mapsto {}^A T_B = \begin{bmatrix} {}^A R_B & {}^A \mathbf{t}_B \\ \mathbf{0} & 1 \end{bmatrix}$
- Simple algebra

$$\begin{aligned} {}^A \mathbf{p} &= {}^A \xi_B \cdot {}^B \mathbf{p} &\mapsto & {}^A \tilde{\mathbf{p}} = {}^A T_B {}^B \tilde{\mathbf{p}} \\ {}^A \xi_C &= {}^A \xi_B \oplus {}^B \xi_C &\mapsto & {}^A T_C = {}^A T_B {}^B T_C \\ \ominus {}^A \xi_B &&\mapsto & {}^A T_B^{-1} \end{aligned}$$

- In the following we will take a closer look at some basic elements of projective geometry that we will encounter when we study the geometrical aspects of imaging
 - Homogeneous coordinates, homogeneous transformations

The projective plane

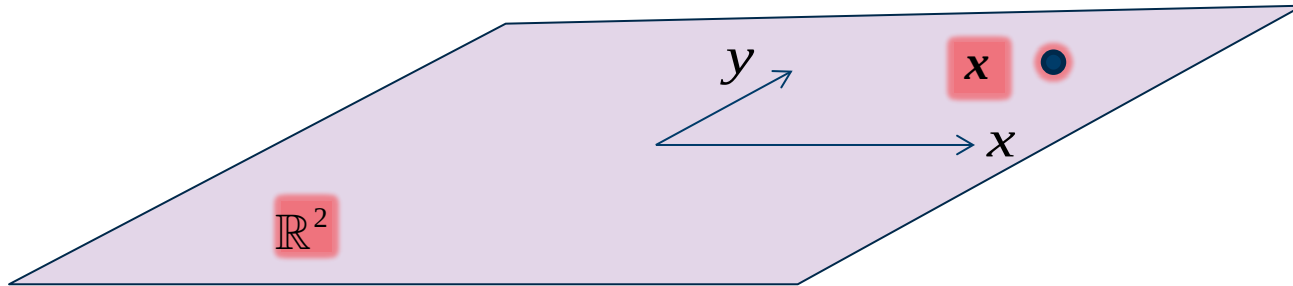
Points

How to describe points in the plane?



The projective plane

Points



How to describe points in the plane?

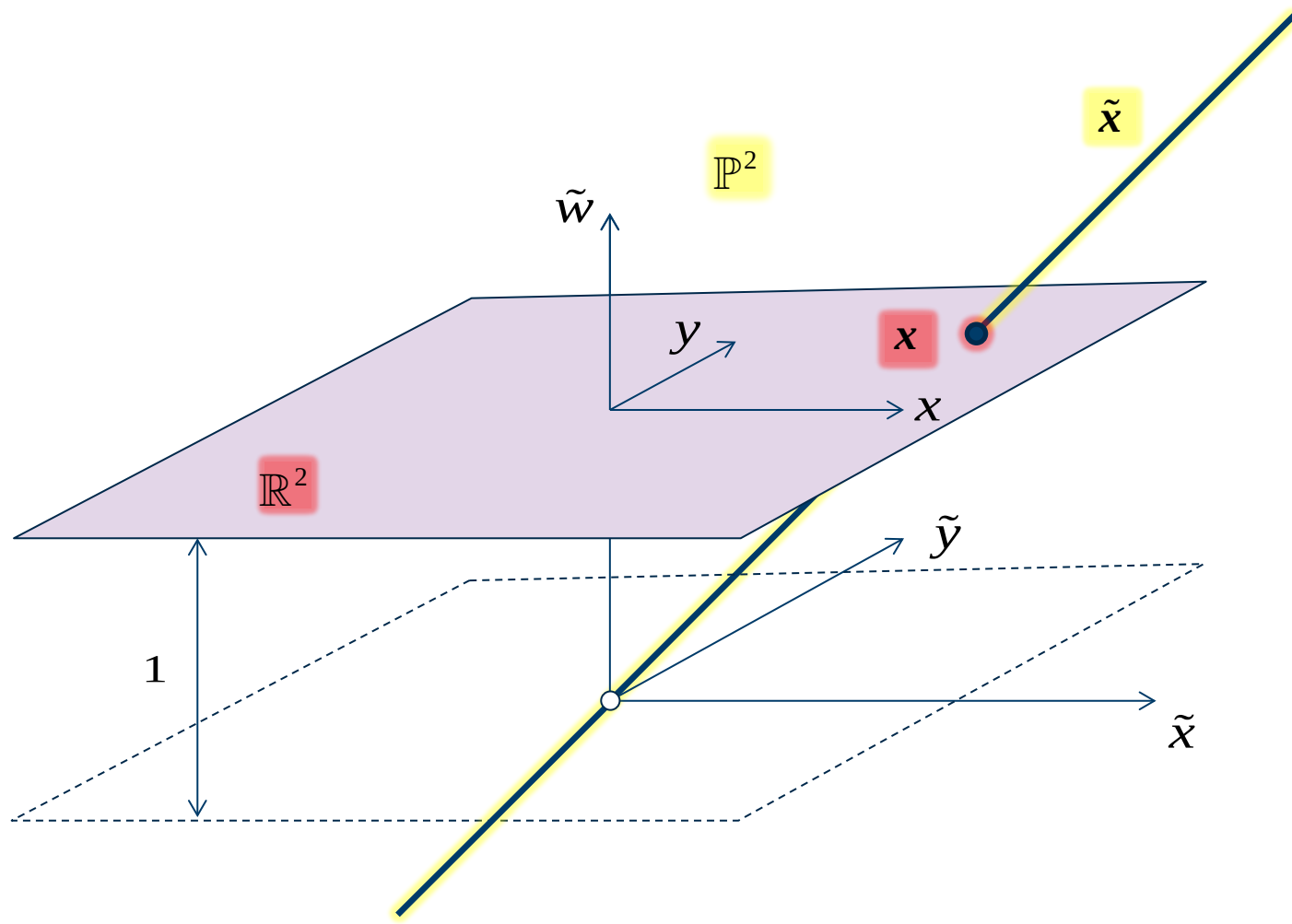
Euclidean plane $\mathbb{R}^2$

- Choose a 2D coordinate frame
- Each point corresponds to a unique pair of Cartesian coordinates

$$\mathbf{x} = (x, y) \in \mathbb{R}^2 \mapsto \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$

The projective plane

Points



How to describe points in the plane?

Euclidean plane $\mathbb{R}^2$

- Choose a 2D coordinate frame
- Each point corresponds to a unique pair of Cartesian coordinates

$$x = (x, y) \in \mathbb{R}^2 \mapsto \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$

Projective plane $\mathbb{P}^2$

- Expand coordinate frame to 3D
- Each point corresponds to a triple of homogeneous coordinates

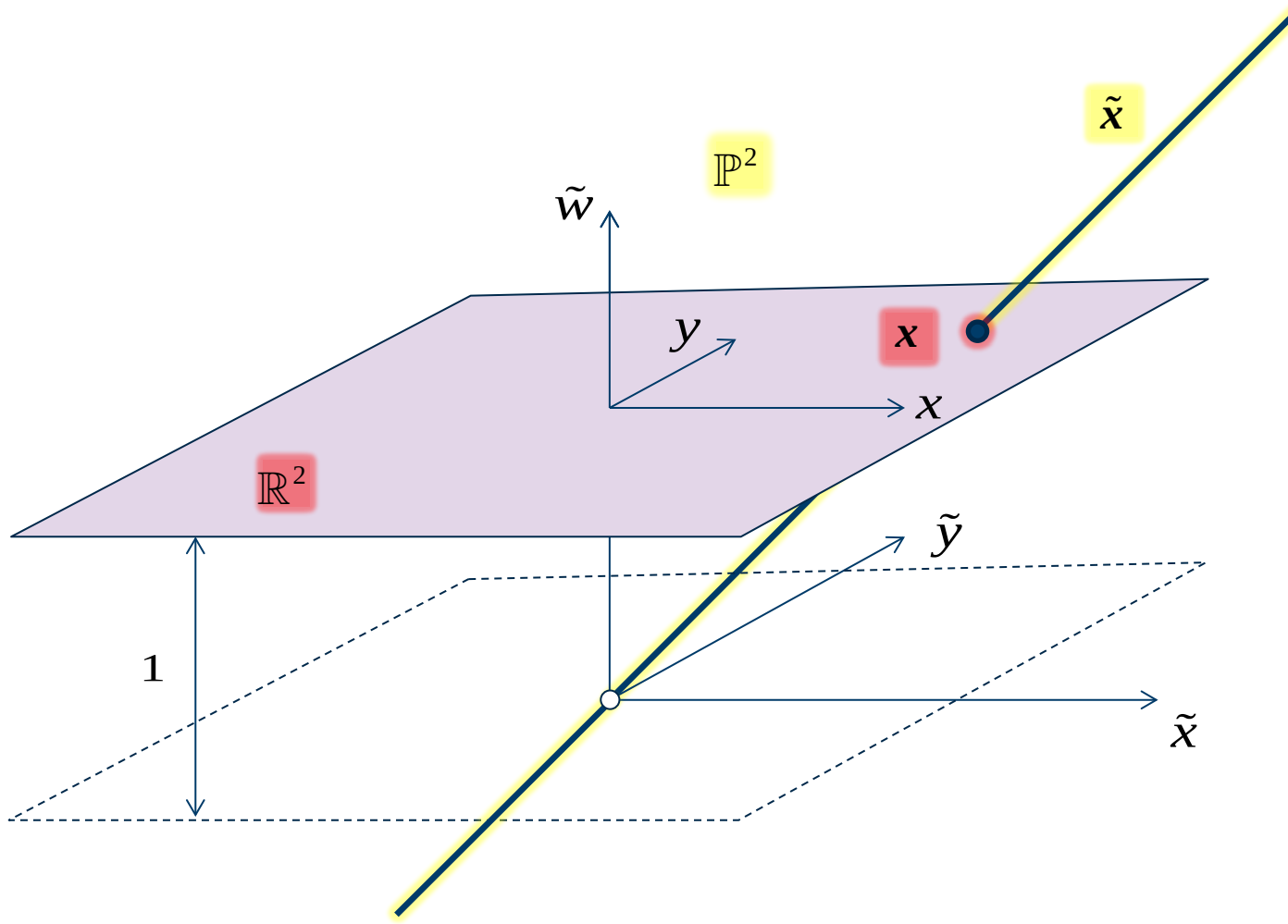
$$\tilde{x} = (\tilde{x}, \tilde{y}, \tilde{w}) \in \mathbb{R}^3 \mapsto \tilde{\mathbf{x}} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{w} \end{bmatrix}$$

s.t.

$$(\tilde{x}, \tilde{y}, \tilde{w}) = \lambda(\tilde{x}, \tilde{y}, \tilde{w}) \quad \forall \lambda \in \mathbb{R} \setminus \{0\}$$

The projective plane

Points



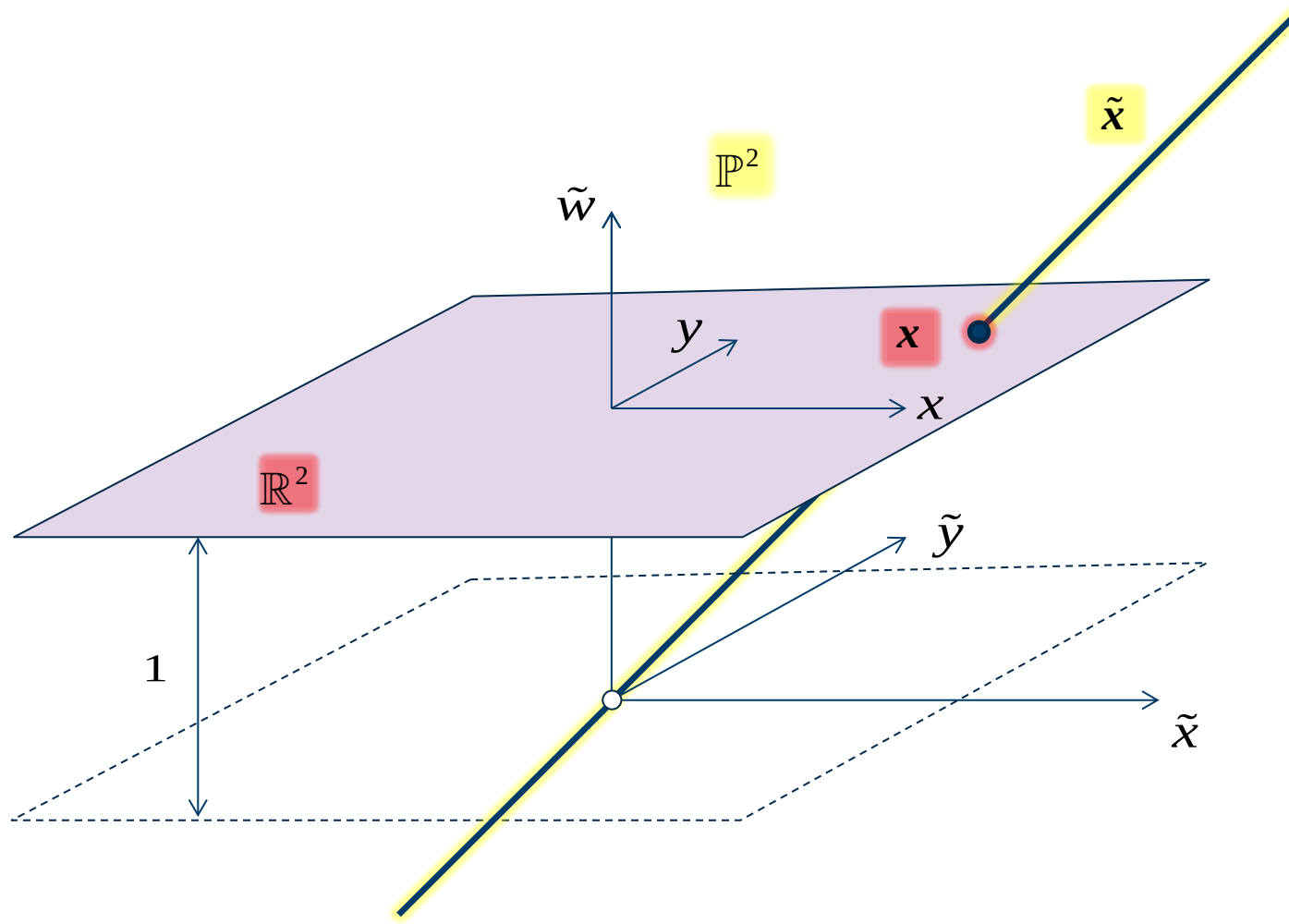
Observations

1. Any point $x = (x, y)$ in the Euclidean plane has a corresponding homogeneous point $\tilde{x} = (x, y, 1)$ in the projective plane
2. Homogeneous points of the form $(\tilde{x}, \tilde{y}, 0)$ does not have counterparts in the Euclidean plane

They correspond to points at infinity and are called *ideal points*

The projective plane

Points



Observations

- When we work with geometrical problems in the plane, we can swap between the Euclidean representation and the projective representation

$$\mathbb{R}^2 \ni \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix} \mapsto \tilde{\mathbf{x}} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \in \mathbb{P}^2$$

$$\mathbb{P}^2 \ni \tilde{\mathbf{x}} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{w} \end{bmatrix} \mapsto \mathbf{x} = \begin{bmatrix} \tilde{x}/\tilde{w} \\ \tilde{y}/\tilde{w} \end{bmatrix} \in \mathbb{R}^2$$

Example

1. These homogeneous vectors are different numerical representations of the same point in the plane

$$\tilde{\mathbf{x}} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix} = \begin{bmatrix} -30 \\ -20 \\ -10 \end{bmatrix} \in \mathbb{P}^2$$

2. The homogeneous point $(1,2,3) \in \mathbb{P}^2$ represents the same point as $\left(\frac{1}{3}, \frac{2}{3}\right) \in \mathbb{R}^2$

The projective plane

Lines

How to describe lines in the plane?



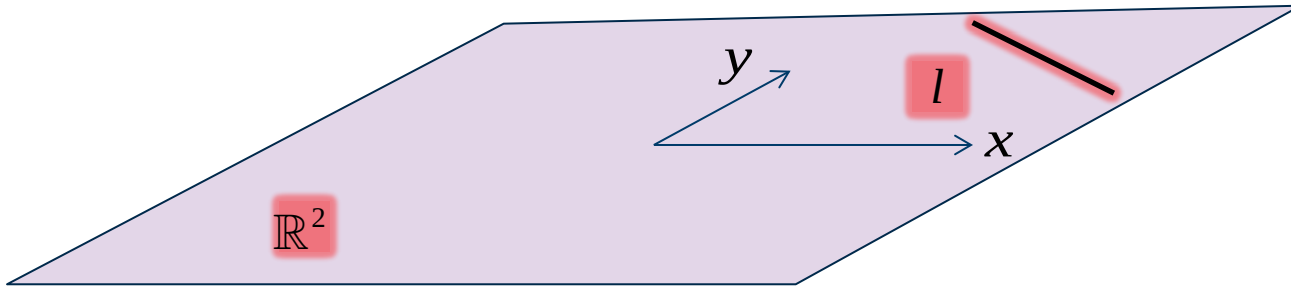
The projective plane

Lines

How to describe lines in the plane?

Euclidean plane $\mathbb{R}^2$

- 3 parameters $a, b, c \in \mathbb{R}$
 $l = \{(x, y) \mid ax + by + c = 0\}$



The projective plane

Lines

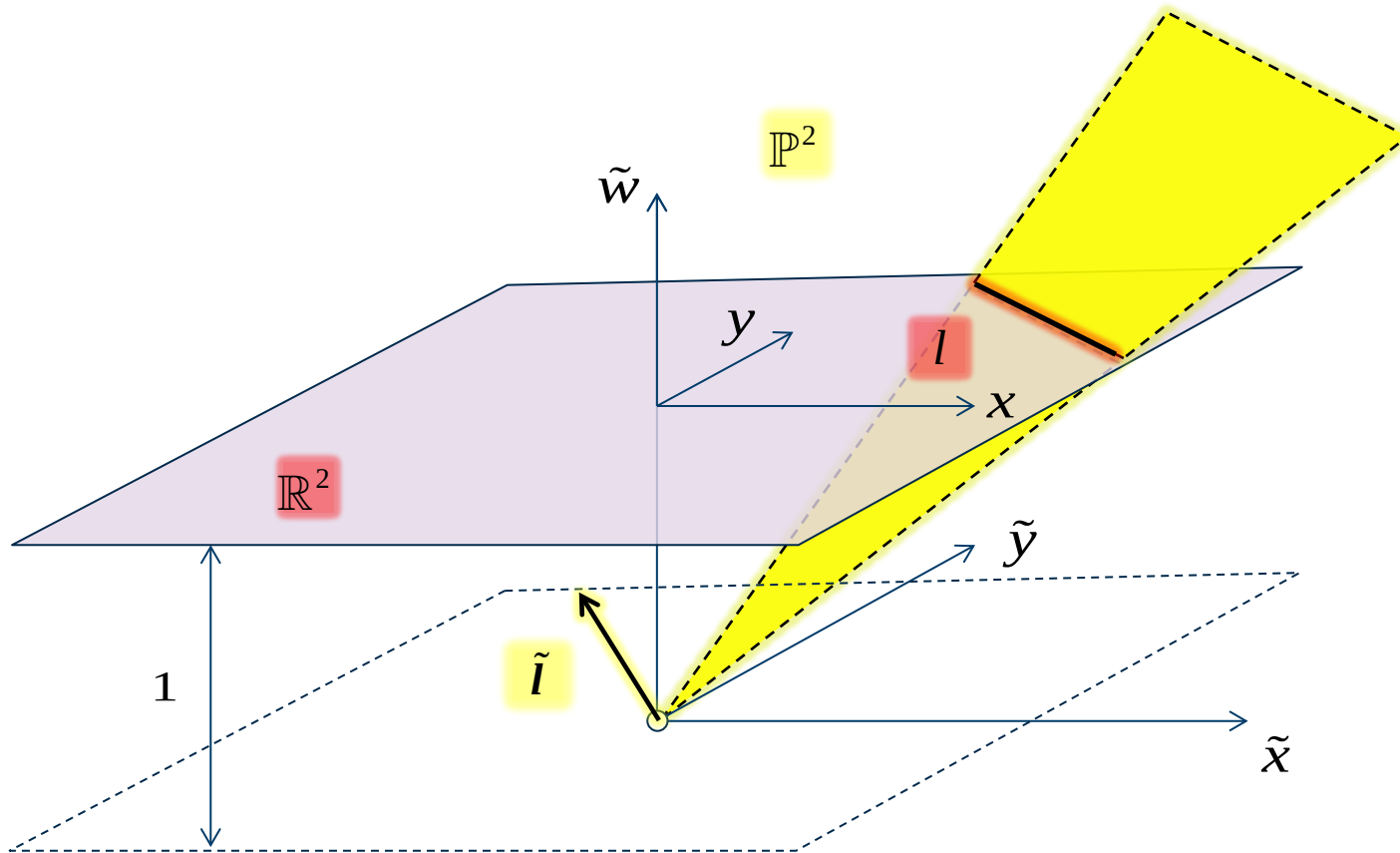
How to describe lines in the plane?

Euclidean plane $\mathbb{R}^2$

- Triple $(a, b, c) \in \mathbb{R}^3 \setminus \{\mathbf{0}\}$
 $l = \{(x, y) \mid ax + by + c = 0\}$

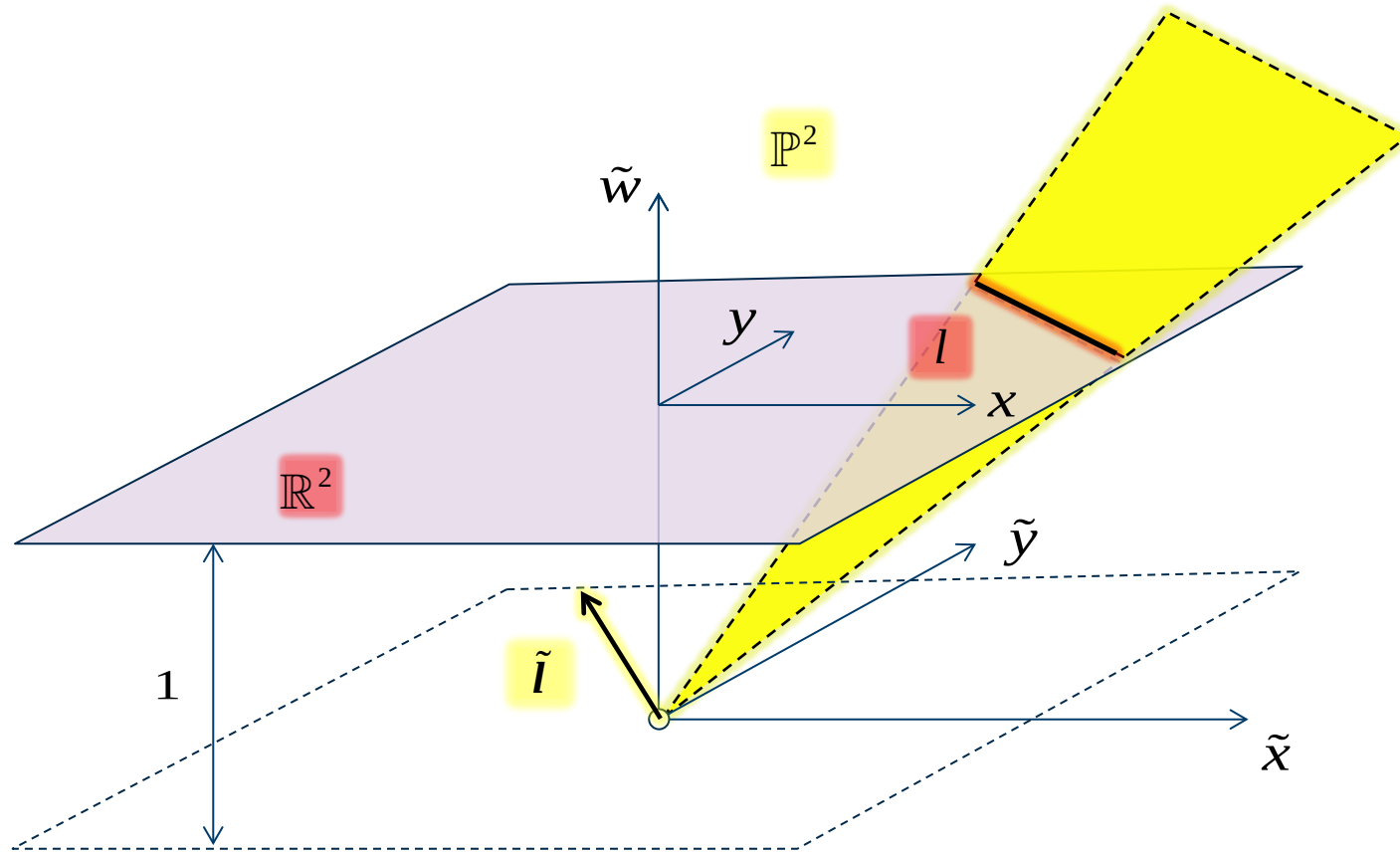
Projective plane $\mathbb{P}^2$

- Homogeneous vector $\tilde{l} = [a, b, c]^T$
 $l = \{\tilde{x} \in \mathbb{P}^2 \mid \tilde{l}^T \tilde{x} = 0\}$



The projective plane

Lines



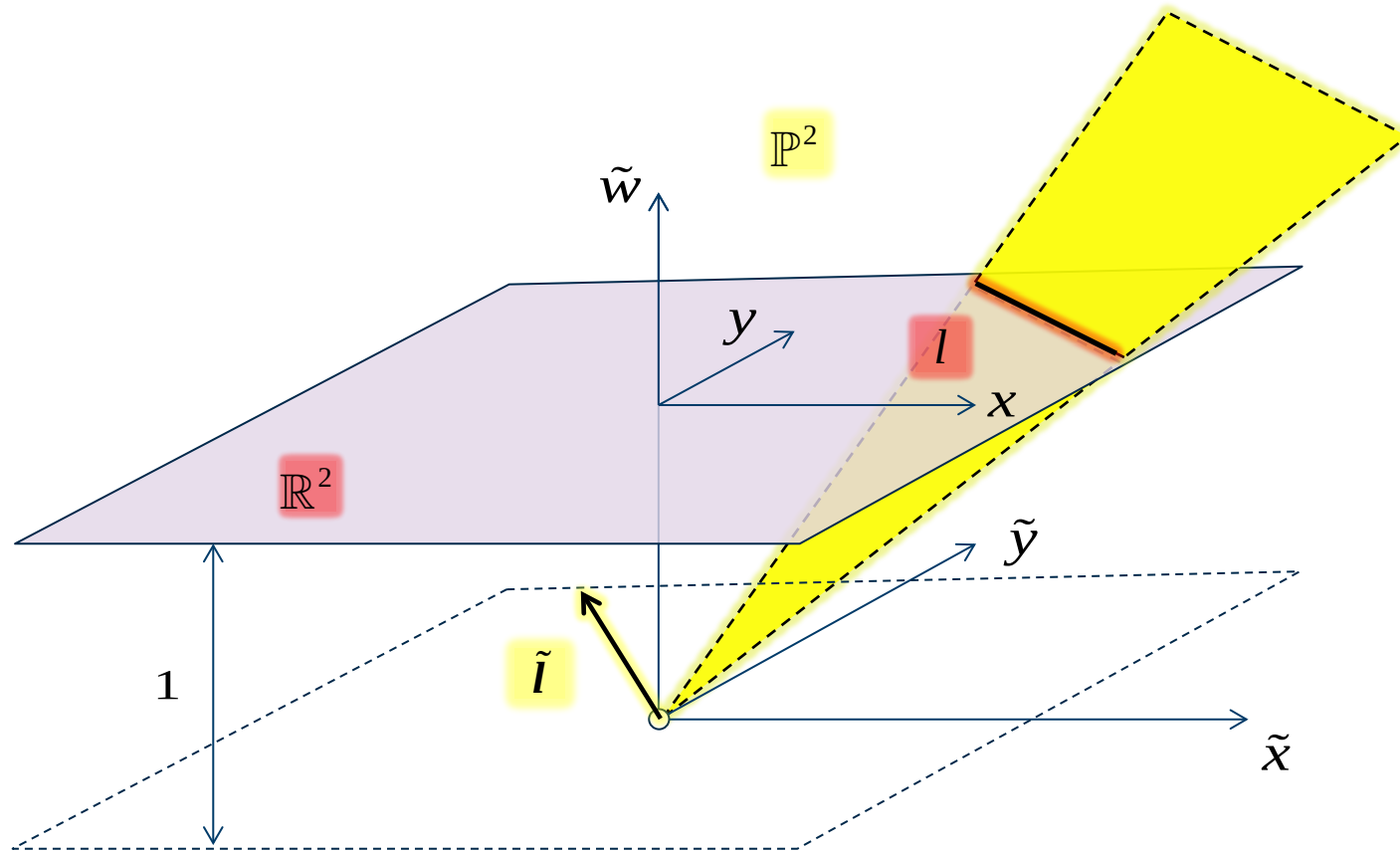
Observations

1. Points and lines in the projective plane have the same representation, we say that points and lines are dual objects in $\mathbb{P}^2$
2. All lines in the Euclidean plane have a corresponding line in the projective plane
3. The line $\tilde{l} = [0,0,1]^T$ in the projective plane does not have an Euclidean counterpart

This line consists entirely of ideal points, and is known as *the line at infinity*

The projective plane

Lines



Properties of lines in the projective plane

1. In the projective plane, all lines intersect, parallel lines intersect at infinity

Two lines $\tilde{l}_1$ and $\tilde{l}_2$ intersect in the point

$$\tilde{x} = \tilde{l}_1 \times \tilde{l}_2$$

2. The line passing through points $\tilde{x}_1$ and $\tilde{x}_2$ is given by

$$\tilde{l} = \tilde{x}_1 \times \tilde{x}_2$$

Example

Determine the line passing through the two points (2, 4) and (5, 13)

Homogeneous representation of points

$$\tilde{\mathbf{x}}_1 = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} \in \mathbb{P}^2 \quad \tilde{\mathbf{x}}_2 = \begin{bmatrix} 5 \\ 13 \\ 1 \end{bmatrix} \in \mathbb{P}^2$$

Homogeneous representation of line

$$\tilde{\mathbf{l}} = \tilde{\mathbf{x}}_1 \times \tilde{\mathbf{x}}_2 = [\tilde{\mathbf{x}}_1]_{\times} \tilde{\mathbf{x}}_2 = \begin{bmatrix} 0 & -1 & 4 \\ 1 & 0 & -2 \\ -4 & 2 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 13 \\ 1 \end{bmatrix} = \begin{bmatrix} -9 \\ 3 \\ 6 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix}$$

Equation of the line

$$-3x + y + 2 = 0 \quad \Leftrightarrow \quad y = 3x - 2$$

Matrix representation
of the cross product

$$\mathbf{u} \times \mathbf{v} \mapsto [\mathbf{u}]_{\times} \mathbf{v}$$

where

$$[\mathbf{u}]_{\times} \stackrel{\text{def}}{=} \begin{bmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{bmatrix}$$

The projective plane

Transformations

- Some important transformations – like the action of a pose ξ on points in the plane – happen to be linear in the projective plane and non-linear in the Euclidean plane
- The most general invertible transformations of the projective plane are known as homographies
 - or projective transformations / linear projective transformations / projectivities / collineations

Definition

A homography of $\mathbb{P}^2$ is a linear transformation on homogeneous vectors represented by a homogeneous, non-singular 3×3 matrix H

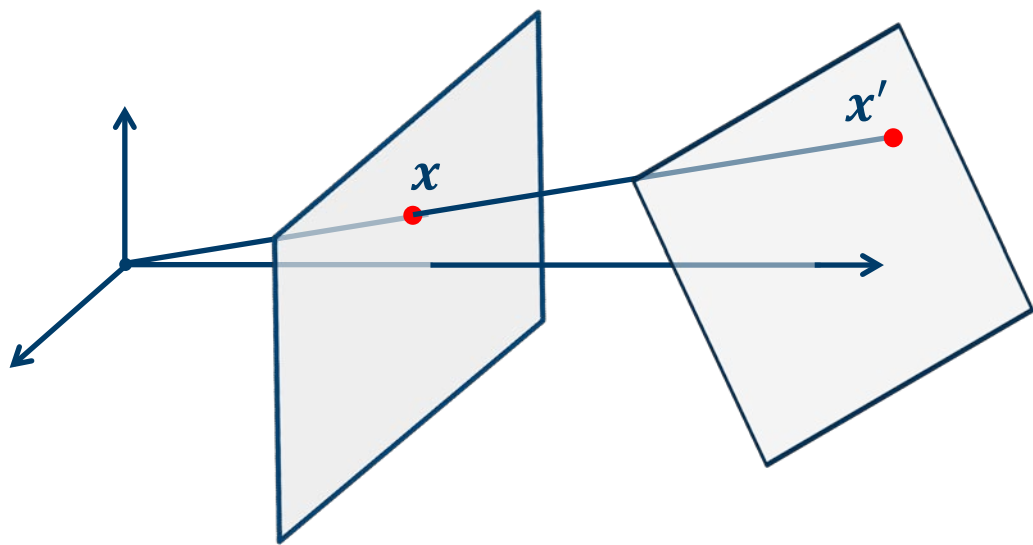
$$\begin{bmatrix} \tilde{x}' \\ \tilde{y}' \\ \tilde{w}' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{w} \end{bmatrix}$$

So H is unique up to scale, i.e. $H = \lambda H \forall \lambda \in \mathbb{R} \setminus \{0\}$

The projective plane

Transformations

- One characteristic of homographies is that they preserve lines, in fact any invertible transformation of $\mathbb{P}^2$ that preserves lines is a homography
- Examples
 - Central projection from one plane to another is a homography
Hence if we take an image with a perspective camera of a flat surface from an angle, we can remove the perspective distortion with a homography



Perspective distortion



Without distortion

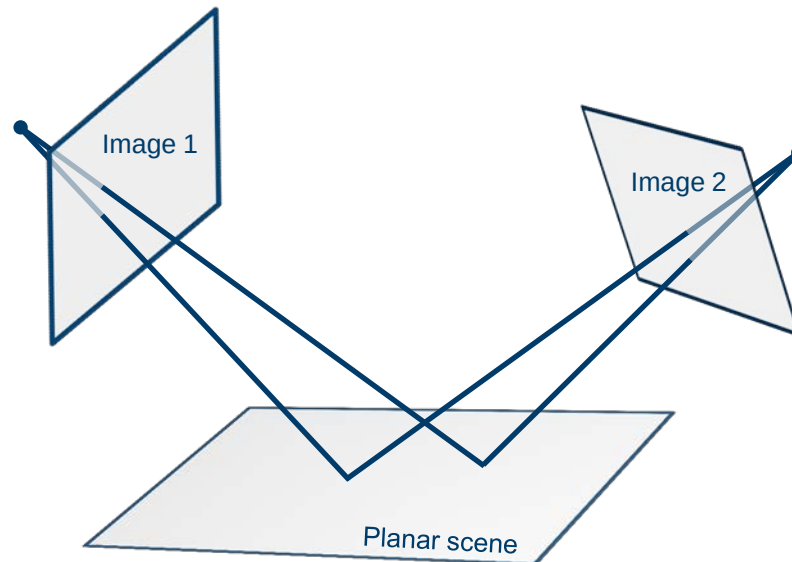


Images from <http://www.robots.ox.ac.uk/~vgg/hzbook.html>

The projective plane

Transformations

- One characteristic of homographies is that they preserve lines, in fact any invertible transformation of $\mathbb{P}^2$ that preserves lines is a homography
- Examples
 - Central projection from one plane to another is a homography
 - Two images, captured by perspective cameras, of the same planar scene is related by a homography



The projective plane

Transformations

- One characteristic of homographies is that they preserve lines, in fact any invertible transformation of $\mathbb{P}^2$ that preserves lines is a homography
- Examples
 - Central projection from one plane to another is a homography
 - Two images, captured by perspective cameras, of the same planar scene is related by a homography
- One can show that the product of two homographies also must be a homography
We say that the homographies constitute a group – the projective linear group $PL(3)$
- Within this group there are several more specialized subgroups

Transformations of the projective plane

Transformation of $\mathbb{P}^2$	Matrix	#DoF	Preserves	Visualization
Translation	$\begin{bmatrix} I & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$	2	Orientation + all below	
Euclidean	$\begin{bmatrix} R & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$	3	Lengths + all below	
Similarity	$\begin{bmatrix} sR & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$	4	Angles + all below	
Affine	$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix}$	6	Parallelism, line at infinity + all below	
Homography /projective	$\begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$	8	Straight lines	

The projective space in 3D

- The relationship between the Euclidean space $\mathbb{R}^3$ and the projective space $\mathbb{P}^3$ is much like the relationship between $\mathbb{R}^2$ and $\mathbb{P}^2$

- In the projective space
 - We represent points in homogeneous coordinates

$$\tilde{\mathbf{x}} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} \lambda \tilde{x} \\ \lambda \tilde{y} \\ \lambda \tilde{z} \\ \lambda \tilde{w} \end{bmatrix} \forall \lambda \in \mathbb{R} \setminus \{0\}$$

- Points at infinity have last homogeneous coordinate equal to zero
- Planes and points are dual objects

$$\tilde{\Pi} = \{\tilde{\mathbf{x}} \in \mathbb{P}^3 \mid \tilde{\boldsymbol{\pi}}^T \tilde{\mathbf{x}} = 0\}$$

- The plane at infinity are made up of all points at infinity

$$\begin{aligned} \mathbb{R}^3 \ni \mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} &\mapsto \tilde{\mathbf{x}} = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \in \mathbb{P}^3 \\ \mathbb{P}^3 \ni \tilde{\mathbf{x}} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \\ \tilde{w} \end{bmatrix} &\mapsto \mathbf{x} = \begin{bmatrix} \tilde{x}/\tilde{w} \\ \tilde{y}/\tilde{w} \\ \tilde{z}/\tilde{w} \end{bmatrix} \in \mathbb{R}^3 \end{aligned}$$

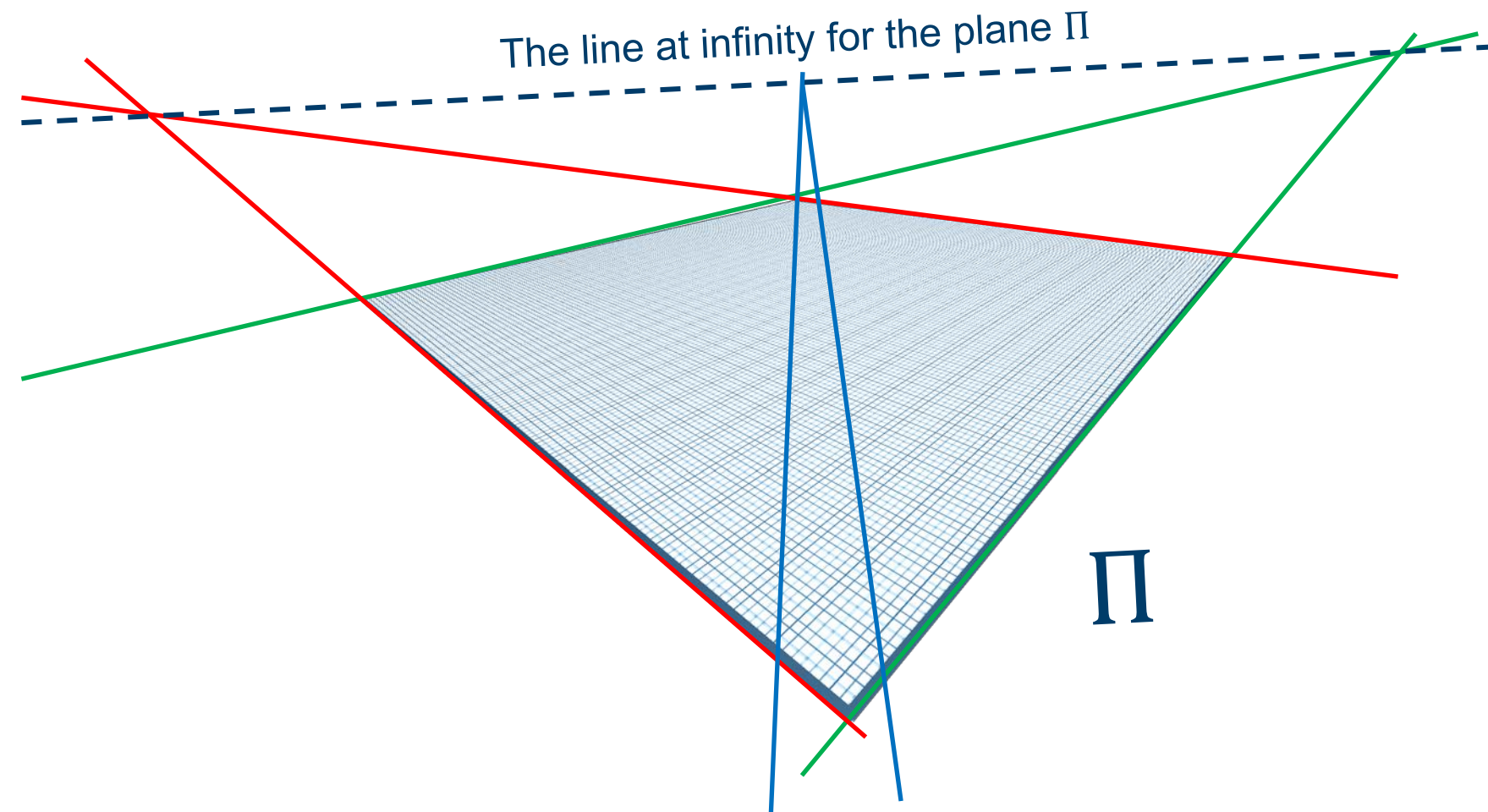
Transformations of the projective space in 3D

Transformation of $\mathbb{P}^3$	Matrix	#DoF	Preserves
Translation	$\begin{bmatrix} I & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$	3	Orientation + all below
Euclidean	$\begin{bmatrix} R & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$	6	Volumes, volume ratios, lengths + all below
Similarity	$\begin{bmatrix} sR & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$	7	Angles + all below
Affine	$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix}$	12	Parallelism of planes, The plane at infinity + all below
Homography /projective	$\begin{bmatrix} h_{11} & h_{12} & h_{13} & h_{14} \\ h_{21} & h_{22} & h_{23} & h_{24} \\ h_{31} & h_{32} & h_{33} & h_{34} \\ h_{41} & h_{42} & h_{43} & h_{44} \end{bmatrix}$	15	Intersection and tangency of surfaces in contact, straight lines

Summary

- The projective plane $\mathbb{P}^2$
 - Homogeneous coordinates
 - Line at infinity
 - Points & lines are dual
- The projective space in 3D $\mathbb{P}^3$
 - Homogeneous coordinates
 - Plane at infinity
 - Points & planes are dual
- Linear transformations of $\mathbb{P}^2$ and $\mathbb{P}^3$
 - Represented by homogeneous matrices
 - Homographies $\supset$ Affine $\supset$ Similarities $\supset$ Euclidean $\supset$ Translations
- Additional reading
 - Szeliski: 2.1.2, 2.1.3,

Example



A point at infinity

